Cluster as a Service

An Obol Offering



Obol DVs - The Gold Standard

When staking ETH, institutions choose Obol Distributed Validators (DVs) — the most resilient, high-performance infrastructure on Ethereum through multi-operator clusters with distributed key generation, threshold signing, and built-in client diversity.

Executive Summary

In September 2025, Kiln's API compromise forced the emergency exit of over 1.6M ETH, spiking the validator exit queue by 150%. This single point of failure cost institutional stakers millions in rewards lost during the lengthy redeployment and reactivation process of those validators. The damage would have been far greater if the compromise had triggered a large slashing event.

Ethereum Validator Queue

ENTRY QUEUE	EXIT QUEUE
21 days	39 days

Data from Oct 19th, 2025. Source: Validator Queue

Obol's Multi-Operator Distributed Validator (DV) clusters eliminate single-operator dependency, ensuring validators remain online and secure even during operator failure.

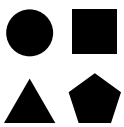
Obol's **Cluster as a Service (CaaS)** for DATs and capital allocators simplifies provisioning and is done in 3 steps.

Step 1. Select your Staking Strategy.
Step 2. Choose your Staking Operators.
Step 3. Deploy with confidence.

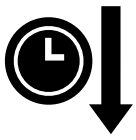
We handle cluster coordination, lifecycle management, monitoring, and performance SLAs end-to-end so you get "always-on" validators with client diversity, zero-downtime, and slashing resistance baked in.



Operator Rotation



Client Diversity



Zero Downtime

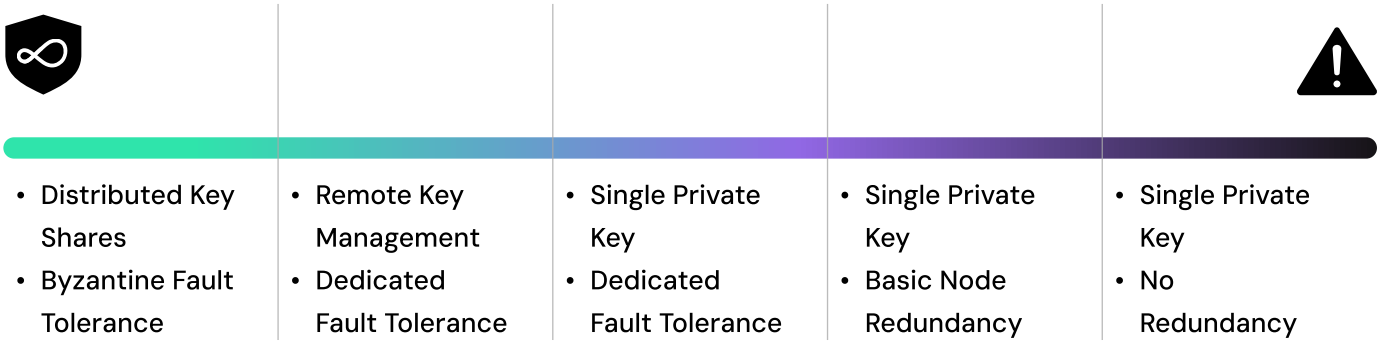
Diversification Using Superior Technology

Capital allocators understand the need for diversifying staking providers. Today, treasury companies spread Ethereum validators across 2-3 operators for safety. Even Coinbase Prime—who used to only offer their own staking infrastructure with Coinbase Cloud—now provides multiple operator options.

Limitations of Legacy Validators

But even when diversifying across a few operators, you face limitations:

- Each operator gets full validator keys, posing a security risk *(Multiple employees have access to validator keys. Only a single one needs to be phished to risk your stake being slashed)*
- You manage multiple deposits, multiple relationships, multiple SLAs
- Rotating operators means exiting or consolidating your validators and potentially waiting weeks in queues
- Vendor lock-in reduces your negotiating power



• Distributed Key Shares • Byzantine Fault Tolerance	• Remote Key Management • Dedicated Fault Tolerance	• Single Private Key • Dedicated Fault Tolerance	• Single Private Key • Basic Node Redundancy	• Single Private Key • No Redundancy

How Multi-Operator Distributed Validators Fix This

- **Distributed Operations** - No single operator holds complete validator keys. Each operator gets cryptographic key shares. Key shares mitigate the risk of key theft; if one operator gets compromised, the other operators keep your validators safe.
- **Single Deposit, Multiple Operators** - Instead of rebalancing operators stake, keys activated in a DV cluster are shared equally amongst the operators, and payments (if you choose) can run programmatically . That means easier activation and exits of validator keys, no juggling inflows and outflows, and lower risk from human handoffs.
- **Your Negotiating Power** - Less vendor lock-in means better pricing. Operators know you can replace them easily, so they compete harder for your business.
- **Operator Rotation Without Exits** - Bad operator performance? Rotate out the operator, not your entire stake. No exit queue. No lost rewards. The Kiln crisis would have been handled in hours, not months, saving millions of dollars in lost rewards.
- **Staking Strategy Agnostic** - Capital allocators can do native staking, liquid staking, or liquid restaking - Obol serves all staking strategies as it sits at the lowest layer of the Ethereum validator stack.

Multi-Operator DVs vs. Legacy Validators

Scenario	Single Operator Legacy Validator	Multi-Operator Distributed Validator
Operator Compromise	Emergency exit all validators (45+ day queue)	Continue operating with existing healthy operators and replace compromised operator
Infrastructure Failure	Complete service disruption	Seamless continuity with majority consensus
Regulatory Risk / Counterparty Issues	Forced unstaking of entire position	Rotate to compliant operators without needing to exit e.g. ability to rotate to operators in new jurisdictions
Performance Degradation	Struggle with poor performance and exit to replace vendor	Replace underperforming operator easily

Industry Leading Risk Adjusted Returns

	Legacy Validators	OBOL DVs
Raver Score	97.35%	**98.63%**
Inclusion Delay	1.023	**1.010**
Uptime	99.8%	**100%**

30d performance on rated.network as of October 6th, 2025

LIVE ON ETHEREUM MAINNET	SECURING OVER	SLASHING INCIDENTS
34+ months	**$3 billion in ETH**	**0**

Cluster as a Service

Step 1: Select your staking strategy

Step 1 in the CaaS process is selecting a staking strategy. Your staking strategy will evolve. Obol DVs sit at the lowest layer of Ethereum's staking stack and don't directly interact with smart contracts—meaning they support any staking strategy, from conservative native staking to yield-optimized restaking, all while maintaining multi-operator security.

Native Staking

w/ Multi-Operator DVs

- Deploy a DV cluster with three or more unique operators (each controlling one node)
- Maximizes yield through OBOL token incentives + optional EigenLayer restaking rewards
- Complete setup and ongoing management: monitoring, operator coordination, deployment
- Easy migration path to other staking options (including Lido v3 stVaults and Liquid Collective SMS)

Liquid Staking

w/ Multi-Operator DVs

- Use DVs within liquid staking protocols (Lido stVaults, Liquid Collective SMS) for LST liquidity
- Combines deep liquidity with your operator choice and compliance requirements
- Optimizes yield through Obol incentives + DeFi strategies while maintaining liquidity optionality
- Full flexibility to choose compliant operators within the LST framework

Restaking and Liquid Restaking

w/ Multi-Operator DVs

- Deploy DVs through liquid restaking protocols (Ether.fi)
- Access restaking rewards from multiple AVS networks
- Maintain multi-operator security while capturing restaking yield premiums
- Liquid restaking tokens provide additional DeFi utility and exit liquidity

Custom Bespoke Staking Solutions

w/ Multi-Operator DVs

Obol operates at the infrastructure layer, so you can select virtually any advanced validator strategy. Pre-confirmation protocols, validator sidecars, MEV routing, and specific builder selection are all on the table. We'll map each option to your objectives, outline risks, and help to quantify potential upside.

Step 2: Choose your staking operators

Operator selection is a critical step in the CaaS process. Obol maintains deep ties with top operators around the globe and packages them into fault-tolerant DV clusters. This choice becomes a strategic lever: diversify by jurisdiction, drive price competition, and rotate operators without exiting validators. Our partners remain the benchmark for institutional-grade operations.

US Partner Node Operators

US

Figment · BLOCKDAEMON · Validation Cloud · ParaFi Tech · galaxy

*All operators maintain US-domiciled infrastructure and US regulatory compliance

International Partner Node Operators

INTL

galaxy · Bitcoin Suisse · SOLST/CE STAKING · BLOCKDAEMON · Figment

DSRV · PIERTWO · Stakely · HASHKEY CLOUD

The operators listed here are not exhaustive. Choose from Ethereum core client teams or from over 200+ independent operators currently running Obol DVs on mainnet. If desired, your organization can participate directly, run one or more nodes in the cluster while partnering with other vetted operators. Obol will curate, onboard, and monitor whichever mix you choose.

Operator Failure: Financial Impact Analysis

Operator failure has a real financial impact. A security compromise can force validators into the exit queue, **where they're still at risk of slashing**. Opportunity cost is compounded during the formal exit and the time spent re-entering the validator set. **Distributed validators reduce this exposure:** if one operator fails, the validator remains online, and operators can be rotated without interrupting service—preserving yield and protecting principal.



Cost of Single-Operator Failure (based on Kiln incident)

Emergency Exit Queue

45+ Days spent in the exit queue. If keys were compromised, this is leverage for the bad actor.

Lost Rewards

Inactivity during the re-creation and re-entry queue will result in significant lost rewards at scale (minimum of 2 weeks of lost rewards). Since March 24, Ethereum is capped to activating a maximum of ~57,000 ETH per day.

Operational Nightmare

Emergency procedures, stakeholder communications, and reputational damage control are tangible second order effects of an operator failure.

Multi-Operator DVT Investment



DVT Fee Premium

Cost is roughly 50 basis points above legacy staking setups. That premium funds redundant, fault tolerant operations, independent jurisdictions, operator selection and rotation, and 24/7 alerting.



Payback Period

Moving from a single operator to DVs pays for itself once the downtime you avoid is worth more than the extra DVT cost. Avoiding 13.5 hours of downtime covers the entire yearly DVT fee. Preventing a 45-day redeployment would cover roughly 80 years of DVT fees.



Built in Monitoring

Obol provides real-time monitoring across DV clusters and operators, issuing actionable alerts before performance degrades. This proactive system reduces mean time to detect, prevent, and resolve incidents.

ROI Calculation (Example 2M ETH Position)

Annual Staking Rewards	~$300M (3% APY)
Daily Reward Generation	~$800k
Obol Fee	~$450K annually
Payback Period	13.5 hours of avoided downtime

Obol Incentives

Obol delivers additional rewards through a fully funded, long-term incentive program for validators using our Distributed Validator technology.

The program is designed to increase staking returns for users of Obol software, ensuring long-term alignment by giving direct exposure to the native token of the future industry standard and end game Ethereum staking technology.

High Upside

Were the Obol token to appreciate in value, Operators would receive a higher APR Boost. As an example:

- *At an FDV of $250M (OBOL $0.50), the Incentive Program would yield an estimated 10% APR boost (~3.3% APR)*
- *At an FDV of $500M (OBOL $1.00), the Incentive Program would yield an estimated 20% APR boost (~3.6% APR)*

How it works

- Tokens are distributed proportionally across all qualified validators in the program.
- The only requirements are staking on Obol DVs and directing the Obol Fee to the Obol Association.
- Partners can claim Obol Incentives weekly through flexible options tailored to their specific needs.

Strategic Value

- **Immediate Staking Return Boost:** Day-one yield boost through token rewards
- **Ecosystem Upside:** Early participation in core Ethereum infrastructure growth
- **ETH per Share Optimization**: Additional rewards help improve ETH per share metrics, driving value to shareholders.

Feature	OBOL Program
APR Boost	Currently ~5% boost (~0.15% APR)
Duration	Designed to last at least 3 years
Token Allocation	Fixed: 12.5M OBOL tokens per year
Distribution	Weekly
Token Issuance	Fully Funded No new issuance
Program Allocation and Supply Impact	Fixed allocation for sustainability
Summary	**Certainty around duration of incentive program with uncapped APR upside**

Insurance and Risk Management

Multi-Layer Insurance Protection

With DVs, built-in redundancy and slashing resistance reduces the likelihood of a catastrophic event. Add in multi-layer insurance coverage (operator policies + a DV-tailored master policy) to address the remaining tail risks, giving board members measurable, contract-based assurance not available to single-operator setups.

Built-In Coverage: Each operator brings their own insurance policy against slashing risk.
Galaxy Exclusive Policy: Market-leading coverage specifically designed for multi-operator distributed validator setups, covering:

- Client-related slashing events
- Operator-related slashing and liveness failures
- Coverage exclusive to DV infrastructure (not available for single-operator setups)

Multiple insurance layers provide superior protection compared to legacy single-operator setups.

Why Act Now

Market Timing

- **Post-Kiln Incident:** Market awareness of operator risk at all-time high
- **Regulatory Clarity:** Institutional tailwinds for Ethereum staking
- **Scale Advantage:** DATs size provides negotiating leverage for premium terms with node operators.

Competitive Differentiation

- **First Mover:** Be the first major DAT to implement multi-operator architecture
- **Risk Leadership:** Demonstrate superior risk management to institutional investors
- **Yield Optimization:** Maintain consistent staking returns regardless of operator issues

In the meantime if you have any questions,
don't hesitate to chat with:

business@obol.tech **Obol.org**